

Lecture 1

Monetary and Fiscal Policy, and Hysteresis

Bank of Portugal—Empirical Macroeconomics

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London School of Economics, Autumn 2025

Introductions

The Sequence

- Structure of the course

- Topics (e.g. wage and price inflation)

- What might be an interesting question

- Tools

- Narrow: shift-share instruments, local projections/VARs, event studies

- Broad: regional data for aggregate questions, identification + natural experiments

- How to answer the question

Lectures

- 1 core paper which you all read in advance (“required reading”). I present.
- Please interrupt and ask questions as much as possible! There are no bad questions

Advice (PhD Students)

- Read Eric Zwick's guide to grad school
- Apply for data early + go hunting for natural experiments
 - Takes 1.5 - 2 years to go from a data idea to a clean set of results
 - Economics has changed—big datasets + creative empirics are important
 - PhD students have an advantage versus faculty:
 - You can spend months at a time working in far flung locations
- Great data in Portugal
 - Employment, banking etc
- Talk to people in other fields and attend their seminars
 - E.g. I went to the labor seminar during grad school
 - Asset pricing, IO, trade, public finance ...

Overview

- Identification in macro
- Monetary policy
- Fiscal policy
- Hysteresis

Identification in Macro

State of Empirical Macro

- Key empirical questions same as 80 years ago:
 - What are the sources of business cycles?
 - How does monetary and fiscal policy affect the economy?
- Why do we not know the answers to these questions?
- Crucial reason: identification in macro is hard

Monetary Policy in the Great Recession

- Monetary policy is endogenous
 - Not just a little endogenous
 - Fed employs hundreds of PhD economists to make policy as endogenous as possible
- Fed lowered rates aggressively in 2008
 - Did so for a reason! (e.g., housing collapse and financial crisis)
 - OLS regression of output on interest rates does not capture effects of monetary policy
 - It captures combined effects of monetary policy and factors causing Fed to act

Two Strands of Empirical Work in Macro

1. Causal inference

- Identify plausibly exogenous variation in some policy (a “natural experiment”)
- Regress outcomes of interest on exogenous policy variation

2. More structural modes of inference

- Use a set of moments to discriminate between models
- GMM / SMM / MLE estimation of a structural model

- Both can be combined fruitfully

Causal Inference in Macro

Two challenges to causal inference:

1. Convincing natural experiments few and far between
2. Rarely see exactly the experiment we want (external validity)

Revolution of Identification in Applied Micro

- Rising standards for what constitutes credible identification (e.g., Angrist-Pischke 10)
- Increased use of:
 - Instrumental variables
 - Difference-in-difference / event studies
 - Regression discontinuities
 - Randomized controlled trials

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 - Go to labor seminars etc.
- How can these methods be used in macro?
- Key challenge:
 - Object estimated often does not correspond directly to object of interest
 - I.e., policy response or structural parameter

External Validity in Macro

1. Term structure of shocks heterogeneous
 - Some only affect short run
 - Some affect short and long run
 - Some only affect longer run (e.g., when monetary policy at ZLB)
 - Responses to these are (very) different in standard models
 2. Fiscal shock depend on monetary response (and vice versa)
 - Multiplier in normal times
 - Multiplier when monetary policy is at ZLB
 3. Policy response depends on state of the economy
 - Degree of slack in the economy
 - How open the economy is
- Amongst other challenges ...
 - One response: gather direct evidence on every different case
 - May not be feasible

Causal Effects as Identified Moments

- Responses to **identified** shocks (i.e., causal effects) can be used as **moments** in structural estimation exercises to help discriminate between models
- Favored structural model then used to answer question of interest
- Terminology: **identified moment** as a shorthand for a target statistic (i.e., **moment**) obtained as a response to an **identified** structural shock (i.e., causal effect)
 - Causal effects often obtained from natural experiments

Example: Marginal Propensity to Consume

- Comes from natural experiments (e.g. Johnson-Parker-Souleles 2006, Parker et al 2013)
 - Limited external validity
 - Studies purely temporary income shock (but most shocks are somewhat persistent)

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 - Studies purely temporary income shock (but most shocks are somewhat persistent)
- Used to discipline structural models of consumption:
 - Simple complete markets models can't match estimates of MPC
 - Standard incomplete market models (e.g. Aiyagari/Bewley/Huggett) have a hard time
 - Kaplan-Violante 14: incomplete markets + illiquid assets can match MPC

Why Use Identified Moments?

- After all, it's a lot of effort!
 - Could only study simpler moments (e.g. mean, variance, covariance)
- Simpler moments typically sensitive to all structural shocks in model
 - May have to make hard-to-verify assumptions all structural shocks
- Identified moments estimate parameters relating to particular shock
 - Identified moments can be invariant to relative variance of other shocks
 - In some cases identified moments independent from parameters in other “blocks” of the model (these can be treated as nuisance parameters)

Example: Marginal Propensity to Consume

- Why not use a simpler moment to estimate consumption models?
 - Could we match the raw correlation between income and consumption growth?
 - Problem: sensitive to additional assumptions about structural shocks and labor supply
 - E.g. what share of income shocks are temporary or permanent?
 - Hard to know in the data, have to make strong assumptions
- MPC of transitory income shock discriminates between consumption models
 - Regardless of whether most income shocks are permanent or temporary

Simpler Moments Still Useful

Simpler moments still sometimes highly informative for disciplining models:

- Micro moments
 - e.g. frequency of price change (Bils & Klenow)
- Macro moments
 - e.g. equity premium (Mehra & Prescott)

The art is knowing when identified moments are particularly useful

Regional Data and Identified Moments

- Increasingly common to estimate identified moments from regional data:
 - Mian-Sufi 14, Nakamura-Steinsson 14, Autor-Dorn-Hanson 13
 - Convincing natural experiments are more common in regional data
 - Supreme Court Justice Louis Brandeis: *"States are the laboratories of democracy"*
- Key challenge—"Missing Intercept"
 - Cross-sectional responses don't directly answer key aggregate questions
 - GE effects absorbed by time fixed effects
- Approach:
 - Use cross-sectional responses as moments in estimation of structural models
 - Use favored structural model to answer aggregate questions
 - We will revisit in this lecture in the context of fiscal policy

Monetary Policy

Effects of Monetary Policy?

- Central question in macroeconomics:
 1. Monetary policy is a central macroeconomic policy tool
 2. Answer helps distinguish between competing models of the world

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- Central question in macroeconomics:
 1. Monetary policy is a central macroeconomic policy tool
 2. Answer helps distinguish between competing models of the world
- Consensus within mainstream U.S. media that effects are large
- No consensus in many other countries
- Until relatively recently: much controversy in academia

Why Don't We Already Know? The Empirical Challenge

Given central importance, how can we not already know?

- **Changes in monetary policy occur for a reason!!**
- Purpose of central banks to conduct systematic policy that reacts to developments in economy

Types of Evidence

- Controlling for Confounding Factors
 - Structural Vector Autoregressions
 - Romer and Romer (2004)
- Discontinuity-Based Evidence / High-Frequency Evidence

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- Controlling for Confounding Factors
 - Structural Vector Autoregressions
 - Romer and Romer (2004)
- Discontinuity-Based Evidence / High-Frequency Evidence
- Also:
 - Narrative evidence (e.g. Friedman & Schwarz 1963)
 - Natural experiments (e.g. Chodorow-Reich et al 2020; Palma 2021)

Dynamic Causal Inference

Objective:

- Causal effect of change in monetary policy at time t on output / prices / etc. at time $t + j$

Two steps:

1. Identify shocks—exogenous variation in (say) monetary policy, maybe using a VAR
2. Estimate effects of shocks on output / prices / etc.

Important to consider these two steps separately

Identification with VARs

- Consider a VAR

$$Y_t = GY_{t-1} + R\varepsilon_t$$

Y_t is a vector of macro variables (e.g. $Y_t = \{i_t, \Delta y_t, \pi_t\}$)

- How do we identify the response of Y_{t+j} to a structural shock ε_t ?

Identification with VARs

- Consider a VAR

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- How do we identify the response of Y_{t+j} to a structural shock ε_t ?
- Suppose you estimate a VAR (i.e., estimate n OLS regressions)
 - You will get:

$$Y_t = GY_{t-1} + u_t$$

where u_t are reduced form errors with variance-covariance matrix Σ

- Unfortunately, Σ not enough to identify R
- **Structural** VARs make additional assumptions to be able to identify R
 - Another way of thinking about it: identification of structural shocks ε_t

SVAR Identification of Monetary Shocks

- Common approach:

- Regress fed funds rate on output, inflation, etc. + a few lags of fed funds rate, output, inflation, etc.

$$i_t = \alpha + \phi_y y_t + \phi_\pi \pi_t + [\text{four lags of } i_t, y_t, \pi_t] + \varepsilon_t$$

- View residual as exogenous variation in monetary policy
- I.e. residual variation due to monetary actions **independent** from endogenous reaction of Fed to state of the economy

- Implies we can identify R in VAR $Y_t = GY_{t-1} + R\varepsilon_t, Y_t = \{i_t, \Delta y_t, \pi_t\}$

- Equivalent to “Cholesky decomposition” / “recursive identification scheme”
- Ordering fed funds rate last

SVARs: Identifying the Shocks

$$i_t = \alpha + \phi_y y_t + \phi_\pi \pi_t + [\text{four lags of } i_t, y_t, \pi_t] + \varepsilon_t$$

What can go wrong?

1. Reverse causation:

- Assumption being made: Correlation between i_t and (π_t, y_t) is due to (π_t, y_t) influencing i_t but not the other way around
 - If i_t influences (π_t, y_t) (contemporaneously), we have a “simultaneous equation problem” (ε_t correlated with (π_t, y_t))
 - Assumption being made: i_t is “fast-moving” variable, while π_t and y_t are slow moving. So i_t doesn't affect π_t and y_t contemporaneously
-
- Often discussion stops here, seems like an innocuous assumption, but is it?

SVARs: Identifying the Shocks

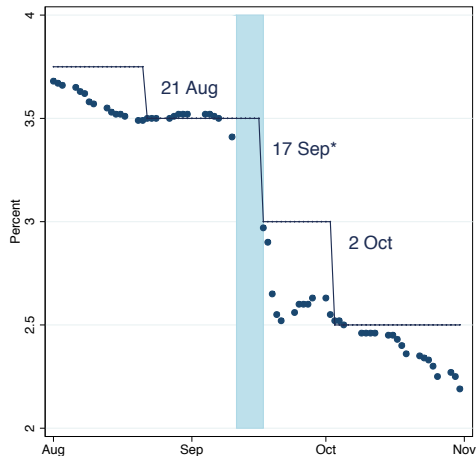
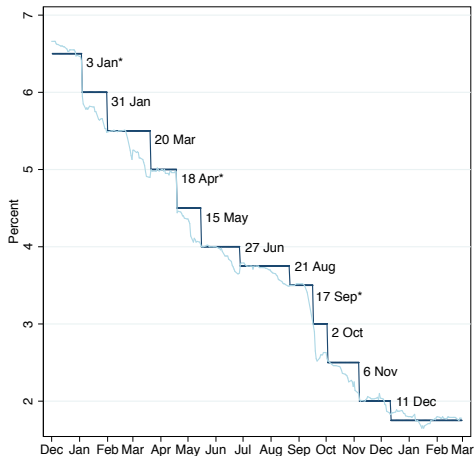
$$i_t = \alpha + \phi_y y_t + \phi_\pi \pi_t + [\text{four lags of } i_t, y_t, \pi_t, \text{etc.}] + \varepsilon_t$$

What can go wrong?

2. Omitted variables bias

- There may be other variables that affect i_t and also y_{t+j} (remember, the goal is to estimate the effect of ε_t on y_{t+j})
- Fed bases policy on huge amount of data
 - Banking sector, stock market, foreign developments, commodity prices, terrorist attacks, temporary investment tax credit, Y2K, etc., etc.
- Too many variables to include in regression!
- Any information used by Fed and not sufficiently controlled for by included controls will result in endogenous variation in policy being viewed as exogenous shock to policy → “anticipation effects”

Anticipation Effects: Was 9/11 a Monetary Shock?



Dark line: Fed funds target. Light line/dots: 1-month eurodollar rate. * indicates unscheduled meeting.

Sample period: Dec 2000 - Feb 2002. Source: Nakamura and Steinsson (2018)

Anticipation Effects: Was 9/11 a Monetary Shock?

- According to structural VARs: Yes!?!
 - Nothing had yet happened to controls in VAR
 - Drop in rates cannot be explained, therefore an exogenous shock
- In reality: Obviously not!
 - Fed dropped rates in Sept 2001 in response to terrorist attack, which affected Fed's assessment of future output growth and inflation
- Any unusual (from perspective of VAR) weakness in output growth after 9/11, perversely, attributed to exogenous easing of monetary policy
- Highly problematic

News Shocks and VARs

- 9/11 an example of a news shock
 - Almost nothing happened to contemporaneous output
 - But event contains news about future output
- Why not just include fast moving variables like stock/bond prices in interest rate equation to capture news?
 - Only makes sense if these variables not affected by contemporary monetary policy
 - But that is clearly not the case
 - Post-treatment controls (endogenous or “bad” controls)

Romer-Romer 04

- Hopeless to control individually for everything in Fed's information set
 - Alternative approach: control for Fed's own forecasts (Greenbook forecasts)
- Key idea:
 - Endogeneity of monetary policy comes from **one thing only**: What Fed thinks will happen to the economy
 - Controlling for this is sufficient

Controlling for Greenbook Forecast

- Romer-Romer's specification:

$$\begin{aligned}\Delta ff_m = & \alpha + \beta ffb_m + \sum_{i=-1}^2 \gamma_i \Delta \tilde{y}_{mi} + \sum_{i=-1}^2 \lambda_i (\Delta \tilde{y}_{mi} - \Delta \tilde{y}_{m-1,i}) \\ & + \sum_{i=-1}^2 \phi_i \tilde{\pi}_{mi} + \sum_{i=-1}^2 \theta_i (\tilde{\pi}_{mi} - \tilde{\pi}_{m-1,i}) + \rho \tilde{u}_{m0} + \varepsilon_m\end{aligned}$$

- Δff_m change in intended FFR at meeting
 - ffb_m level before meeting
 - $\tilde{y}$, $\tilde{\pi}$, $\tilde{u}$ forecasts of output, inflation, and unemployment
 - Both forecasts and change in forecasts since last meeting included

Does this Make Sense?

- Residual ε_m considered **exogenous** monetary policy shock
- Does this make sense? **Exogenous to what?**

Romer-Romer 04:

It is important to note that the goal of this regression is not to estimate the Federal Reserve's reaction function as well as possible. What we are trying to do is to purge the intended funds rate series of movements taken in response to useful information about future economic developments. Once we have accomplished this, it is desirable to leave in as much of the remaining variation as possible.

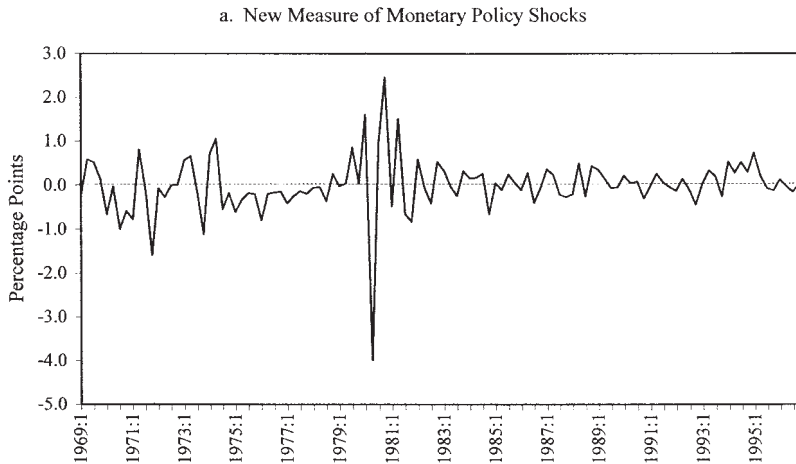
What Is a Monetary Shock?

- Fed does not roll dice
- Every movement in intended fed funds rate is a response to something
- Some are responses to something that directly affects outcome variable of interest
 - These are endogenous
- Reactions to anything else (exchange rate, political pressure, etc)
conditional on output forecast count as a shock

What Are the Shocks?

1. Variation in Fed operating procedure important
 - E.g., emphasis on monetary quantities in 1979-1982
2. Variation in policy makers' beliefs about workings of economy
 - In early 1970's Fed believed inflation highly unresponsive to slack (Romer-Romer 02)
3. Variation in policy maker preferences/goals
 - E.g., time-varying distaste for inflation
4. Political influences
 - E.g., Arthur Burns set loose policy in 1977 to get re-appointed
5. Pursuit of other objectives
 - At some times, Fed concerned about exchange rate

Romer-Romer Shocks



Source: Romer and Romer (2004).

Potential Issues with Romer-Romer Identification

- Results are highly sensitive to inclusion of Volcker period (Coibion 2012)
 - Excluding Volcker lowers size of effects
- Do we believe the identification scheme?
 - Greenbook forecast has to control for **all information available to Fed**
 - Fed policymakers may have additional information (e.g. 9/11)
- Common view from practitioners: “anything goes”

What Do We Do with These Shocks?

- Dynamic causal inference involves two steps:
 1. Identifying exogenous variation in policy (the shocks)
 2. **Estimating an impulse response given the shocks**

- Three methods to construct impulse response:
 1. Directly regress variable of interest on shock (Jorda 05)
 2. Iterate forward VAR
 3. Iterate forward univariate AR specification (Romer-Romer 04)

Direct Regressions – Jorda Specification

- Simple approach: Regress variable of interest directly on shock: (perhaps including some pre-treatment controls)

$$y_{t+j} - y_{t-1} = \alpha + \beta v_t + \Gamma X_{t-1} + e_{t+j}$$

- Variable of interest: $y_{t+j} - y_{t-1}$
- Monetary shock: v_t
(Could be identified recursively, by Romer-Romer method, or something else)
- Pre-treatment controls: X_{t-1}
- Separate regression for each horizon j
- Also known as local projection (LP)

VARs and Impulse Response Functions

- Consider a VAR

$$Y_t = GY_{t-1} + R\varepsilon_t$$

here: $Y_t = \{i_t, \Delta y_t, \pi_t, \text{and lags}\}$

- Suppose we are interested in effect of $\varepsilon_{1,0}$ on y_t for $t \geq 0$
(Where $\varepsilon_{1,0}$ is the structural monetary shock, identified recursively or by Romer-Romer)
 - Iterate forward the VAR starting at time 0:

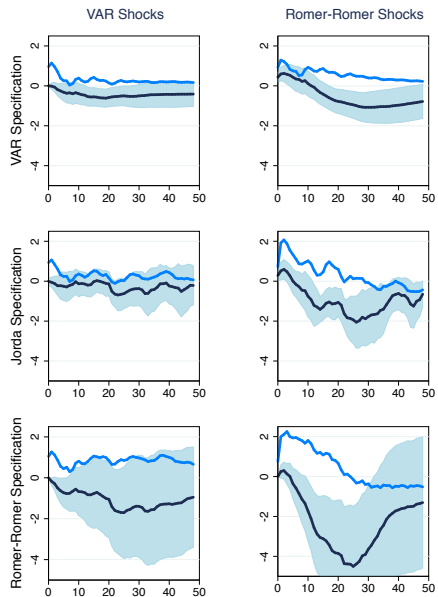
$$Y_t = G^{t-1}R\varepsilon_0$$

- Can estimate G by OLS
- Can estimate R by identification assumptions implied by the structural shocks
(See Ramey 2016 for details)
- Then we can calculate dynamic causal effect of $\varepsilon_{1,0}$ on Δy_t

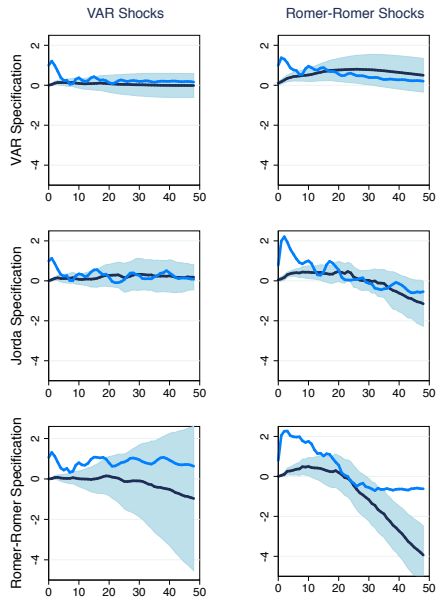
Romer-Romer 04 Impulse Response

$$\Delta y_t = a_0 + \sum_{k=1}^{11} a_k D_{kt} + \sum_{i=1}^{24} b_i \Delta y_{t-i} + \sum_{j=1}^{36} c_j S_{t-j} + e_t$$

- Δy_t monthly change in industrial production
 - D_{kt} month dummies (they use seasonally unadjusted data)
 - S_t monetary shocks
 - Assume money doesn't affect output contemporaneously (No contemporaneous monetary shock)
- Impulse response:
 - Effect on y_{t+1} is c_1
 - Effect on y_{t+2} is $c_1 + (c_2 + b_1 c_1) \dots$



Black line: Industrial production. Blue line: Real interest rate, right panel is Romer-Romer shocks, left is VAR shocks



Black line: CPI. Blue line: Nominal interest rate, right panel is Romer-Romer shocks, left is VAR shocks

Equivalence between LP and VAR IRFs

- We can construct IRFs by: (1) Local projection (2) VAR
 - In practice this choice makes a great deal of difference!
 - Which should we choose?

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Theorem from Plagborg-Moeller & Wolf (2021):

“We prove that local projections (LPs) and Vector Autoregressions (VARs) estimate the same impulse responses. This nonparametric result only requires unrestricted lag structures.”

Equivalence between LP and VAR IRFs

- Plagborg-Moeller & Wolf (2021) result holds:
 - In population
 - With enough lags included and (all the same variables in the regression)
 - Even if the true data generating process is nonlinear
(In these circumstances Romer-Romer method will also estimate the same IRF)
- Intuition:
 - VARs with long lag lengths capture all the linear comovements between y_{t+j} and Y_t
 - Local projections (by design) estimate the linear comovement between y_{t+j} and v_t
- Common and wrong belief before Plagborg-Moeller & Wolf (2021)
 - LP more robust to model misspecification and nonlinearity than VAR
(e.g. Ramey 2016)

Similarity between LP and VAR in Finite Samples

- In finite samples:
 - VAR and LP estimate approximately the same IRF to horizon p
 - If the VAR contains p lags
 - But disagree at longer horizons
- Key takeaways:
 - VAR + LP disagree due to finite sample considerations
 - Choice of lag length is crucial
 - Main discrepancy should be at longer horizons

Discrepancy at Longer Horizons

- Li, Plagborg-Moeller & Wolf (2021): **bias-variance tradeoff**
 - LP is unbiased but imprecise at long horizons
 - VAR is more precise but biased
- Intuition:
 - Consider the VAR(p) $Y_t = GY_{t-1} + R\varepsilon_t$ and IRF $Y_{t+h} = G^{t-1}R\varepsilon_t$ where $h > p$
 - G is estimated using information from the first p sample autocovariances
 - But extrapolated to longer horizons
 - Extrapolation typically biases long horizon IRFs towards zero

My Recommendations (Pinch of Salt)

Use local projection as a starting point!

- Easy to implement
- Easy to communicate to non-macroeconomists and combine with applied micro tools
 - E.g. assess pre-trends
- Much easier to compute nonlinear IRFs (Ramey 2016)
- Much easier to calculate standard errors, even if data non-stationary (Plagborg-Moeller & Montiel-Olea 2021)
- Estimating long horizon IRFs without bias is important! If long horizon IRFs are imprecise we are in trouble anyway ...

Types of Evidence on Monetary Policy

- Controlling for Confounding Factors
 - Structural Vector Autoregressions
 - Romer and Romer (2004)
- Discontinuity-Based Evidence / High-Frequency Evidence

Identification in Macro

High Frequency Identification

- Substantial monetary news released at the end of FOMC meeting
- Possible to use a “discontinuity” based identification approach
- Study changes in market interest rates during window around FOMC meeting
 - Narrow window (One-day or 30 minutes)

Main Regression and Benefits for Identification

- Main regression

$$\Delta s_t = \alpha + \gamma \Delta i_t + \varepsilon_t$$

where:

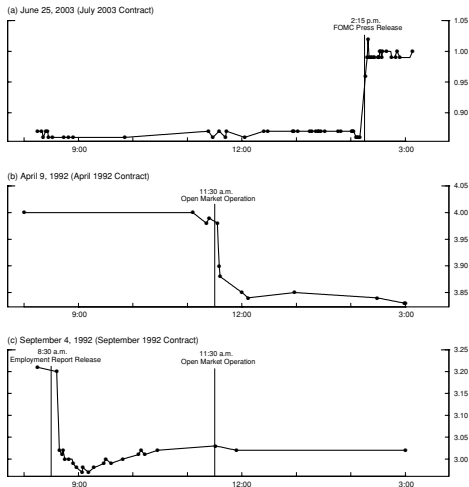
- t is a month
- Δi_t is change in market interest rates in window around monthly FOMC announcement
- Δs_t is change in outcome variable of interest, e.g.:
 - **High frequency:** asset price changes in a narrow window around FOMC announcements
 - **Lower frequency:** monthly changes in GDP during the announcement month
- Main advantage: **rules out anticipation effects**
 - E.g. suppose that before FOMC announcement there is bad news (e.g. 9/11)
 - Bad news should be “baked in” to interest rates before FOMC announcement
 - Δi_t should be uncorrelated with bad news

Threats to Identification

1. If there are other shocks during window:
 - Policy indicator will be contaminated by these shocks because Fed may respond
 - These same shocks may directly affect future variables
 - No longer estimating a causal effect of monetary shocks
2. If entire response of interest rates doesn't occur in narrow window:
 - Estimate of monetary shock biased because shock size biased
 - Might be over-reaction or under-reaction

Key Question: How long should the window be?

Figure 1. Intraday Trading in Federal Funds Futures Contracts



The Power Problem

- Effects on output and inflation are imprecise
- Conventional wisdom—shocks are small:
 - High frequency method leaves out lots of shocks
 - All news about monetary policy on non-FOMC days not captured
 - Sample period is short (only back to late 1980's)
 - Outcomes are noisy
 - Many other shocks affect output and inflation over a 1 year horizon

(I am not convinced this logic is correct—see below)

The Power Problem

Potential solution:

- Focus on outcome variables that move **contemporaneously**
 - E.g., real yields and forwards (from TIPS) (Hanson-Stein 15, Nakamura-Steinsson 18)
 - Essentially a discontinuity based identification strategy

Results from Nakamura-Steinsson (2018)

TABLE 1

Response of Interest Rates and Inflation to the Policy News Shock

	Nominal	Real	Inflation
2Y Treasury Yield	1.10 (0.33)	1.06 (0.24)	0.04 (0.18)
5Y Treasury Yield	0.73 (0.20)	0.64 (0.15)	0.09 (0.11)
10Y Treasury Yield	0.38 (0.17)	0.44 (0.13)	-0.06 (0.08)
2Y Treasury Inst. Forward Rate	1.14 (0.46)	0.99 (0.29)	0.15 (0.23)
3Y Treasury Inst. Forward Rate	0.82 (0.43)	0.88 (0.32)	-0.06 (0.15)
5Y Treasury Inst. Forward Rate	0.26 (0.19)	0.47 (0.17)	-0.21 (0.08)
10Y Treasury Inst. Forward Rate	-0.08 (0.18)	0.12 (0.12)	-0.20 (0.09)

Large Effects on Real Rates

Main take-away:

- Nominal and real rates move one-for-one several years out into term structure
- Response of break-even inflation is delayed and small

Consistent with large effects of monetary policy (lots of nominal rigidity)

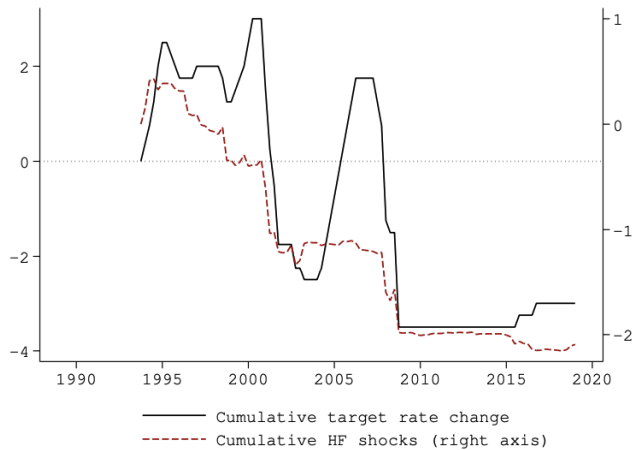
- Transmission of real interest rates to output relatively uncontroversial
- See also Gertler & Karadi (2015)

Potential Issues with This Strategy

- **Problem #1:** shocks are not that small, so why is the effect on output so noisy?
- **Problem #2:** information effect—Fed decision also transmits information to real economy
- **Problem #3:** asset pricing—lots of strange things happen around Fed announcement windows
- **Problem #4:** these shocks are serially correlated!

Problem #1: Shocks Are Not That Small

Figure 5: Evolution of Federal Funds Target and Shocks



Source: Koren, Liu, Mian & Sufi (2021)

Problem #2: Information Effect

TABLE III
RESPONSE OF EXPECTED OUTPUT GROWTH OVER THE NEXT YEAR

	1995–2014	2000–2014	2000–2007	1995–2000
Policy news shock	1.01 (0.32)	1.04 (0.35)	0.95 (0.32)	0.79 (0.63)
Observations	120	90	52	30

Notes. We regress changes from one month to the next in survey expectations about output growth over the next year from the Blue Chip Economic Indicators on the policy news shock that occurs in that month (except that we drop policy news shocks that occur in the first week of the month because we do not know whether these occurred before or after the survey response). Specifically, the dependent variable is the change in the average forecasted value of output growth over the next three quarters (the maximum horizon over which forecasts are available for the full sample). See Online Appendix F for details. We present results for four sample periods. The longest sample period we have data for is 1995m1–2014m4; this is also the period for which the policy news shocks are constructed. We also present results for 2000m1–2014m4 (which corresponds to the sample period used in Table I), 2000m1–2007m12 (a precrisis sample period), and 1995m1–1999m12. As in our other analysis, we drop data from July 2008 through June 2009. Robust standard errors are in parentheses.

Problem #3: Asset Pricing

- Hillenbrand (2022): *“in this paper I document a striking fact: a narrow window around Fed meetings fully captures the secular decline in U.S. Treasury yields since 1980. By contrast, yield movements outside this window are transitory and wash out over time.”*
- Lucca & Moench (2015): *“we document large average excess returns on U.S. equities in anticipation of monetary policy decisions made at scheduled meetings of the Federal Open Market Committee (FOMC) in the past few decades. These pre-FOMC returns have increased over time and account for sizable fractions of total annual realized stock returns.”*

Problem #4: Serial Correlation

Ramey (2016):

*I then conducted some further investigations of the Gertler-Karadi shock. Several features emerge. First, the shock is not zero mean. The mean is 0.013 and is statistically different from zero. Second, it is serially correlated; if I regress it on its lagged value, the coefficient is 0.31 with a robust standard error of 0.11. **This is not a good feature since it is supposed to capture only unanticipated movements in interest rates.***

Monetary Policy—Taking Stock

- Identification Strategies
 - Controlling for confounding factors
 - Discontinuity based evidence
- Methods: VARs vs. local projections
- Lessons: Leading identification strategies have flaws
 - Empirical macro is hard!
 - Lots of work to be done even on basic questions

Fiscal Policy

Fiscal Stimulus in Theory—Summary

Neoclassical models:

- Positive but small multipliers (less than one)
 - Due to negative wealth effect
 - Hours $\uparrow$ consumption $\downarrow$

New Keynesian models:

- Multiplier depends on monetary policy response:
 - Constant real rate—multiplier 1
 - Lean against the wind—multiplier < 1
 - Zero lower bound—multiplier > 1
- Multipliers larger with credit constrained agents
- No single aggregate multiplier!

What's The Right Regression?

- Government purchases multiplier:

- When government purchases go up by \$1, how many dollars does output increase by?

$$Y_t - Y_{t-1} = \alpha + \beta (G_t - G_{t-1}) + \varepsilon_t$$

- Usually divide through by Y_{t-1}

$$\frac{Y_t - Y_{t-1}}{Y_{t-1}} = \alpha + \beta \left(\frac{G_t - G_{t-1}}{Y_{t-1}} \right) + \varepsilon_t$$

- Why?

- Reduces heteroskedasticity

- Many other challenges for “estimating the right regression” (Ramey & Zubairy 2018)

Empirical Challenges

$$\frac{Y_t - Y_{t-1}}{Y_{t-1}} = \alpha + \beta \left(\frac{G_t - G_{t-1}}{Y_{t-1}} \right) + \varepsilon_t$$

Identification challenges?

Empirical Challenges

$$\frac{Y_t - Y_{t-1}}{Y_{t-1}} = \alpha + \beta \left(\frac{G_t - G_{t-1}}{Y_{t-1}} \right) + \varepsilon_t$$

Identification challenges?

- Government might spend more when output is low (“automatic stabilisers”)
- Then OLS is downward biased

Fiscal Stimulus: Aggregate Evidence

- Barro-Redlick 11 use evidence from wars:
 - Government purchases multiplier of 0.6-0.7
 - Most identification from WWI, WWII, Korean War
 - Conceptually: Use wars as instrument for spending
 - Potential violations of exclusion restriction?

Fiscal Stimulus: Aggregate Evidence

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 - Government purchases multiplier of 0.6-0.7
 - Most identification from WWI, WWII, Korean War
 - Conceptually: Use wars as instrument for spending
 - Potential violations of exclusion restriction? patriotism, rationing, etc.
- Blanchard-Perotti 02 use SVAR:
 - Peak output response of 1.3 after 15 quarters
 - Very large standard errors
 - Highly sensitive to sample period, controls (see, e.g., Gali et al. 07, Ramey 11, 16)

Fiscal Stimulus: Aggregate Evidence

- Additional weakness of evidence:
 - Highly sensitive to monetary reaction
- Monetary reaction to fiscal shock:
 - Normal time: “leans against the wind”
 - At ZLB: Not able to lean against the wind
- Aggregate multiplier may be very different at ZLB (See Ramey-Zubairy 18, Miyamoto et al. 17 for direct evidence)
- Telling apart RBC model and NK model crucial
 - Both can yield multipliers around 0.7 in normal times
 - But NK model implies much bigger multipliers at ZLB

Fiscal Stimulus: Cross-Sectional Evidence

- Explosion of work post Great Recession:
 - Chodorow-Reich et al. 12, Wilson 12, Shoag 15, Nakamura-Steinsson 14, Acconcia et al. 14, Dupor-Mehkari 16, etc.
- Survey by Chodorow-Reich 17
- Estimates of local fiscal multiplier cluster at 1.5-2.0

Example: Nakamura-Steinsson 14

- Regress output growth on change in military spending at state level the U.S.

$$\left(\frac{Y_{it} - Y_{it-2}}{Y_{it-2}} \right) = \alpha_i + \gamma_t + \beta \left(\frac{G_{it} - G_{it-2}}{Y_{it-2}} \right) + \Gamma X_{it} + \varepsilon_{it}$$

(why divide through by $Y_{i,t-2}$)?

- Basic idea for identification:
 - National military buildups exogenous to **relative conditions** in states receiving disproportionate procurement spending
- Instrument: Fitted value of state spending on national spending allowing different sensitivity for each state
- Year fixed effects (controls for all aggregate shocks)
- State fixed effects (state specific trends)

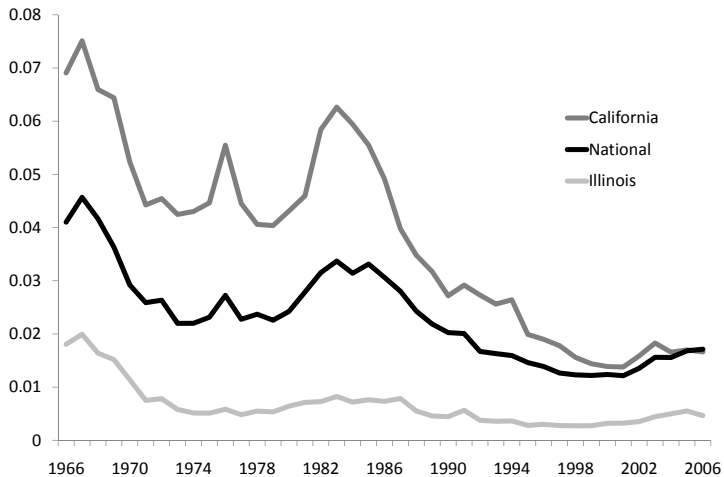


Figure: Prime Military Contracts as a Fraction of State GDP

TABLE I
The Effects of Military Spending

	Output		Output defl. state CPI		Employment		CPI		Pop.
	State	Region	State	Region	State	Region	State	Region	State
Prime Military Contracts	1.43 (0.36)	1.85 (0.58)	1.35 (0.36)	1.91 (0.65)	1.28 (0.29)	1.76 (0.62)	0.03 (0.18)	-0.26 (0.45)	-0.10 (0.17)
Num. Obs.	1989	390	1989	390	1989	390	1785	350	1989

What Do We Learn?

- Local multiplier not the same as aggregate multiplier:
 - States don't have to pay for spending (financed federally)
 - Spillovers to other states
 - Monetary policy doesn't respond in cross-section
 - This is an example of the “missing intercept” problem
- One reaction:
 - Local multiplier estimate not so useful
 - Don't answer the right question (which is aggregate multiplier)
- Different reaction:
 - Perhaps relative multiplier is a powerful statistic in distinguishing between competing models (e.g., RBC vs. New Keynesian)
 - Aggregate multiplier is actually not very strong on that front
 - When know what the right model is we can do good analysis at aggregate level

TABLE 6—GOVERNMENT SPENDING MULTIPLIER IN SEPARABLE PREFERENCES MODEL

	Closed economy aggregate multiplier	Open economy relative multiplier
<i>Panel A. Sticky prices</i>		
Volcker-Greenspan monetary policy	0.20	0.83
Constant real rate	1.00	0.83
Constant nominal rate	∞	0.83
Constant nominal rate ($\rho_g = 0.85$)	1.70	0.90
<i>Panel B. Flexible prices</i>		
Constant income tax rates	0.39	0.43
Balanced budget	0.32	0.43

Notes: The table reports the government spending multiplier for output deflated by the regional CPI for the model presented in the text with the separable preferences specification. Panel A presents results for the model with sticky prices, while panel B presents results for the model with flexible prices. The first three rows differ only in the monetary policy being assumed. The fourth row varies the persistence of the government spending shock relative to the baseline parameter values. The fifth and sixth rows differ only in the tax policy being assumed.

TABLE 7—GOVERNMENT SPENDING MULTIPLIER IN GHH MODEL

	Closed economy aggregate multiplier	Open economy relative multiplier
<i>Panel A. Sticky prices</i>		
Volcker-Greenspan monetary policy	0.12	1.42
Constant real rate	7.00	1.42
Constant nominal rate	∞	1.42
Constant nominal rate ($\rho_g = 0.50$)	8.73	2.04
<i>Panel B. Flexible prices</i>		
Constant income tax rates	0.00	0.30
Balanced budget	-0.18	0.30

Notes: The table reports the government spending multiplier for output deflated by the regional CPI for the model presented in the text with the GHH preferences specification. Panel A presents results for the model with sticky prices, while panel B presents results for the model with flexible prices. The first three rows differ only in the monetary policy being assumed. The fourth row varies the persistence of the government spending shock relative to the baseline parameter values. The fifth and sixth rows differ only in the tax policy being assumed.

What Do We Learn?

- Plain vanilla RBC model inconsistent with regional multiplier
- Plain vanilla NK model inconsistent with regional multiplier
- “Ultra Keynesian” model consistent with regional multiplier
- “Ultra Keynesian” model implies large aggregate effects of fiscal stimulus when monetary policy is constrained

Hysteresis

Motivation

Blanchard, Cerutti & Summers (2015):

“The hysteresis hypothesis ... the hypothesis that recessions may have permanent effects on the level of output relative to trend”

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- Lucas (1988): *“Once one starts to think about [long run growth], it is hard to think about anything else.”*
- But what if business cycles affect long run growth?

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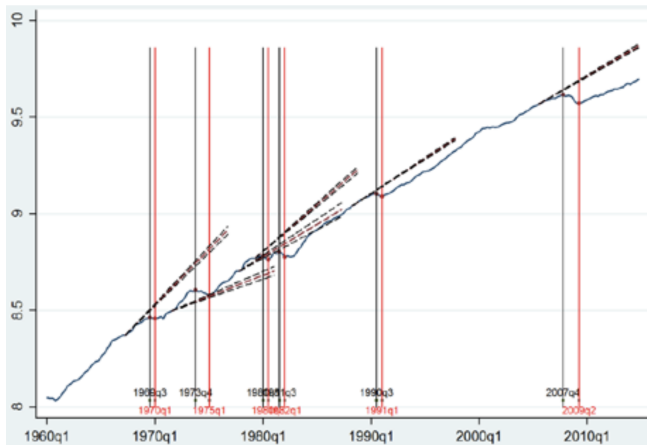
- Lucas (1988): *“Once one starts to think about [long run growth], it is hard to think about anything else.”*
- But what if business cycles affect long run growth?

Great empirical research topic

- Very little high quality empirical work
- Lots of plausible (but largely untested) theoretical mechanisms

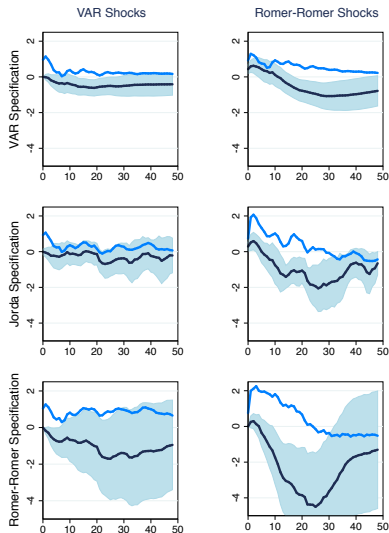
Existing Evidence—Simple Moments

United States: Log Real GDP and Extrapolated Trends



Source: Blanchard, Cerutti & Summers (2015)

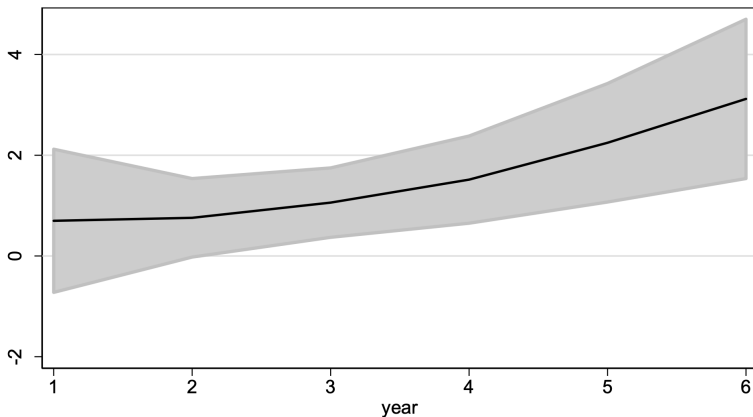
Existing Evidence—Monetary



Black line: Industrial production. Blue line: Real interest rate, right panel is Romer-Romer shocks, left is VAR shocks

Existing Evidence—Fiscal

Long horizon response of output to Nakamura-Steinsson (2014) Fiscal Shock



Source: Ramey discussion of Guren, McKay, Nakamura & Steinsson (2020)

State of the Literature

- Most macroeconomists do not seem to believe in hysteresis—why?
 - Long horizon creates scope for more omitted variables
 - Long horizon estimates tend to be imprecise
 - People's “default model” does not include hysteresis— “natural rate hypothesis”
 - **Need high quality empirics to make progress**
- Many models of hysteresis
 - Labor markets and scarring of unemployed workers (e.g. Pissarides 1992)
 - Technology and endogenous growth (e.g. Benigno & Fornaro 2018)
 - **Need high quality empirics to validate models**

Conclusion

- Identification in macro
- Monetary policy
- Fiscal policy
- Hysteresis
- Reading for next time: Parker, Souleles, Johnson, McClelland (2012)